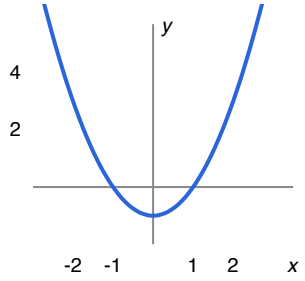
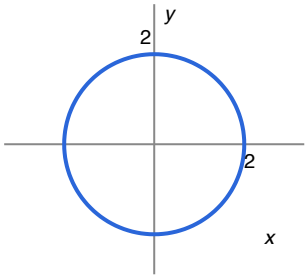
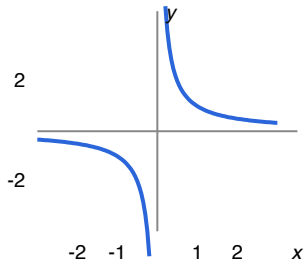
Algebra 8 — Functions · Domain and range · Composite and inverse functions

Marks in brackets. ★ = harder question. $\mathbf{fg}(x)$ means do g first, then f.

1 Decide, with a reason, whether each graph shows a function. [3]







a $y = x^2 - 1$ **b** the circle $x^2 + y^2 = 4$ **c** $y = \dfrac{1}{x}$

> Use the vertical-line test: a graph is a function if no vertical line meets it more than once. **a** Yes — every x gives exactly one y. **b** No — a vertical line through $x = 1$ meets the circle at two points ($y = \pm\sqrt{3}$), so one input has two outputs. **c** Yes — one output for every x in its domain ($x \neq 0$).

2 $\mathbf{f} : x \mapsto x^2 - 2x$. Find, simplifying your answers, [3]

a $\mathbf{f}(3x)$ **b** $3\mathbf{f}(x)$ **c** $\mathbf{f}(-x)$

> **a** Replace x by $3x$: $(3x)^2 - 2(3x) = 9x^2 - 6x$. **b** $3(x^2 - 2x) = 3x^2 - 6x$. **c** $(-x)^2 - 2(-x) = x^2 + 2x$.

3 The domain of $\mathbf{f} : x \mapsto x^2 + 1$ is all real numbers. Write down the range. [2]

> $x^2 \geq 0$ for every real x, so $x^2 + 1 \geq 1$: the range is $\mathbf{f}(x) \geq 1$ (all real numbers ≥ 1).

4 State which values of x (if any) must be excluded from the domain of [3]

a $\mathbf{f}(x) = \sqrt{x + 2}$ **b** $\mathbf{f}(x) = x^3$ **c** $\mathbf{f}(x) = \dfrac{5}{2x - 1}$

> **a** No square roots of negatives: need $x + 2 \geq 0$, so exclude $x < -2$. **b** None — any real number can be cubed. **c** No division by zero: exclude $x = \frac{1}{2}$.

5 $\mathbf{f}(x) = 3x + 1$ and $\mathbf{g}(x) = x - 4$. Find $\mathbf{fg}(5)$. [2]

> g first: $\mathbf{g}(5) = 1$. Then f: $\mathbf{f}(1) = 4$. So $\mathbf{fg}(5) = 4$.

6 $\mathrm{f}(x) = (x-1)^2$ and $\mathrm{g}(x) = 3x$. Find [4]

a $\mathrm{fg}(x)$ **b** $\mathrm{gg}(x)$

> **a** $\mathrm{fg}(x) = \mathrm{f}(3x) = (3x-1)^2$. **b** $\mathrm{gg}(x) = \mathrm{g}(3x) = 9x$.

7 Find the inverse function of [4]

a $\mathrm{f}(x) = \dfrac{x}{3} + 2$ **b** $\mathrm{f}(x) = \dfrac{4}{x+1}$

> Write $y = \ldots$, swap x and y, make y the subject.
>
> **a** $y = \dfrac{x}{3} + 2 \rightarrow x = \dfrac{y}{3} + 2 \rightarrow y = 3(x-2)$. So $\mathrm{f}^{-1}(x) = 3(x-2) = 3x - 6$.
>
> **b** $x = \dfrac{4}{y+1} \rightarrow x(y+1) = 4 \rightarrow y = \dfrac{4}{x} - 1$. So $\mathrm{f}^{-1}(x) = \dfrac{4}{x} - 1$ (or $\dfrac{4-x}{x}$).

8 $\mathrm{f}(x) = 2(x-3)$ and $\mathrm{g}(x) = 2x + 5$. Given that $\mathrm{f}^{-1}(a) + \mathrm{g}^{-1}(a) = 4$, find a. [4]

> Inverses: $\mathrm{f}^{-1}(x) = \dfrac{x}{2} + 3$ and $\mathrm{g}^{-1}(x) = \dfrac{x-5}{2}$. Then
> $\dfrac{a}{2} + 3 + \dfrac{a-5}{2} = 4 \Rightarrow \dfrac{2a-5}{2} + 3 = 4 \Rightarrow 2a - 5 = 2 \Rightarrow a = 3.5$. (Check: $\mathrm{f}^{-1}(3.5) = 4.75$, $\mathrm{g}^{-1}(3.5) = -0.75$, sum 4 ✓.)

Total: 25 marks

Check: function = one output for each input (vertical-line test) · $\mathrm{f}(2x)$: substitute $2x$ for every x; $2\mathrm{f}(x)$: double the whole thing · domain excludes division by 0 and square roots of negatives · $\mathrm{fg}(x)$ = g first · inverse: swap x and y, rearrange.